

Adjuvant Abemaciclib Plus Endocrine Therapy in the Treatment of High-Risk Early Breast Cancer: ASCO Guideline Rapid Recommendation Update Q and A

Rachel A. Freedman, MD, MPH¹; Stephanie L. Graff, MD²; Mark R. Somerfield, PhD³; Melinda L. Telli, MD⁴; Antonio C. Wolff, MD⁵; and Sharon H. Giordano, MD, MPH⁶

In 2020, Johnston et al¹ reported the primary results from monarchE, a phase III, open-label randomized clinical trial, following the second preplanned interim analysis after 19 months of median follow-up. These results showed that the addition of 2 years of abemaciclib, a cyclin-dependent kinase 4/6 inhibitor, to standard adjuvant endocrine therapy (ET) significantly improved invasive-free survival (IDFS) in the intent-to-treat (ITT) population of patients with hormone receptor (HR)-positive, human epidermal growth factor receptor 2 (HER2)-negative, node-positive early breast cancer at high risk of early recurrence. In 2021, Harbeck et al² reported absolute improvements in 3-year IDFS and distant relapse-free survival rates of 5.4% and 4.2%, respectively, for the ITT population in the final prespecified IDFS analysis for study participants receiving abemaciclib. The US Food and Drug Administration (FDA) then approved adjuvant abemaciclib, but limited the approval to the subset of patients with a worse prognostic risk on the basis of a higher anatomic stage (cohort 1) and with higher tumor proliferation based on Ki-67 biomarker expression by immunohistochemistry.

In a Letter to the Editor following the decision by FDA, the investigators reported overall survival (OS) data on the basis of the first interim OS analysis.³ Although the OS data were immature, the observed hazard ratios for OS of 1.091 (95% CI, 0.818 to 1.455) for the ITT population and 0.767 (95% CI, 0.511 to 1.152) in the FDA-approved population appear to have influenced the label indication. In a Rapid Communication to *Journal of Clinical Oncology*,⁴ the FDA offered the following statement regarding the current approval in the United States for adjuvant abemaciclib for a subgroup of the ITT population on the monarchE study: “*While the benefit risk profile was favorable for the indicated subpopulation (ie, statistically significant improvement in invasive disease-free survival [IDFS] for patients with HR+, HER2-negative, node-positive, early breast cancer [cohort 1] with Ki-67 score $\geq$ 20% at the final IDFS analysis with a hazard ratio of 0.643 [95% CI, 0.475 to 0.872; $P = .0042$], given the immaturity and potential OS detriment, it was not favorable for the ITT population.*”

To offer a prompt response to these updated analyses,² ASCO published a rapid recommendation update of ASCO’s selection of optimal adjuvant chemotherapy and targeted therapy guideline⁵ to address the use of abemaciclib in the adjuvant setting (Table 1).⁶ This companion Q and A article addresses some of the questions that clinicians face as they implement monarchE trial results and the rapid recommendation update into clinical practice.

Question: Whom Are You Prescribing Adjuvant Abemaciclib to in Clinical Practice?

ASCO’s rapid update (January 20, 2022) recommended that 2 years of adjuvant abemaciclib plus ET may be offered to patients with HR-positive, HER2-negative, node-positive early breast cancer with a high risk of recurrence, consistent with the monarchE study ITT population inclusion criteria, namely, patients with either ≥ 4 positive axillary lymph nodes or 1-3 positive axillary lymph nodes and either grade 3 disease, tumor size ≥ 5 cm, or Ki-67 index $\geq 20\%$. This recommendation was consistent with the trial design and was broader than the FDA indication, which was limited to a subset of patients from Cohort 1 (≥ 4 nodes or 1-3 nodes if tumor ≥ 5 cm or grade 3) with Ki-67 $\geq 20\%$.

Although we acknowledge the rationale behind the FDA indication, the ASCO rapid recommendation update⁶ suggested broader consideration of abemaciclib for the entire ITT population, given the similar, statistically significant relative reduction in risk of relapse observed with abemaciclib in patients with both Ki-67-high and Ki-67-low tumors (despite different prognostic profiles and differences in absolute benefit for 3-year IDFS—7.1% among Ki-67-high cohort and 4.5% among Ki-67-low population) and because of the challenges in the universal clinical application of Ki-67. As follow-up data mature for patients on monarchE, we will update the ASCO guideline, as appropriate.

Question: How Should We Be Using Ki-67 When Considering Abemaciclib?

At this time, the Ki-67 selection required by the FDA approval excludes a cohort of patients expected to

Author affiliations and support information (if applicable) appear at the end of this article.

Accepted on February 28, 2022 and published at ascopubs.org/journal/op on April 4, 2022; DOI <https://doi.org/10.1200/OP.22.00140>

have a particularly high anatomic risk of recurrence, namely, patients with ≥ 4 lymph nodes involved (N2 or N3 disease). These patients were particularly enriched in the Ki-67-low cohort, with 55% of all patients with ≥ 4 positive lymph nodes in monarchE being Ki-67-low. Although follow-up data remain incomplete for the ITT population, the improvement in IDFS to date was statistically apparent for the entire monarchE ITT population who received abemaciclib, including those with Ki-67-low and Ki-67-high tumors, as above. There are potential challenges in implementing Ki-67 testing into routine clinical practice because of the observed variability and limited standardization in Ki-67 scoring across centers.^{7,8} Consequently, the ASCO rapid recommendation update considered the full ITT population results that showed benefit of abemaciclib across patients with varying Ki-67 scores. We have recommended that adjuvant abemaciclib may be offered per monarchE in those with HR-positive breast cancer with either ≥ 4 positive lymph nodes (regardless of Ki-67 score) or 1-3 lymph nodes with high-risk features (grade 3 disease or tumor ≥ 5 cm or Ki-67 $\geq 20\%$), as per the study's ITT inclusion criteria, recognizing Ki-67 may not be readily available or implemented.

Question: What Is the Optimal Timing for Initiation of Adjuvant Abemaciclib and Can Abemaciclib Be Administered With Radiation?

The monarchE trial enrolled patients after a maximum of 16 months from surgery to random assignment, focusing on those who recently completed active therapy ($> 95\%$ received neo/adjuvant chemotherapy and radiation). monarchE eligibility also allowed up to 12 weeks of ET at the time of random assignment. This timing of abemaciclib initiation in the trial closely aligns with clinical practice, allowing sufficient time for patients to complete recommended adjuvant chemotherapy and radiation.

It is important to note that the monarchE trial does not evaluate the benefit of adding abemaciclib to ET for patients who are longer into their adjuvant ET course and who would not have been eligible for the study. In clinical practice, any administration of abemaciclib should ideally be offered in a similar context and timing to administration in the monarchE study. Patients receiving radiation and chemotherapy on monarchE all completed these treatments before random assignment and treatment with abemaciclib; for these reasons, and for toxicity concerns, all other therapies should be completed, and the only therapy offered concurrently with abemaciclib should be ET.

Question: If a Patient Has *BRCA1* or *BRCA2* Mutation, Can We Combine Abemaciclib and Olaparib in High-Risk Patients?

For patients meeting criteria for monarchE but who also have a germline pathogenic or likely pathogenic variant (PV) in *BRCA1* or *BRCA2*, dual or sequential therapy with olaparib was not examined in the trial. The OlympiA trial⁹ separately

established the significant IDFS benefit of administering one year of adjuvant olaparib after surgery in those with *BRCA1* or *BRCA2* PVs and high-risk disease. Although outcome data are not available for patients in monarchE with germline *BRCA1/2* PVs, the extent of IDFS benefit in OlympiA (8.8% at 3 years) appears higher than that observed in monarchE to date (5.4% in the ITT) and may be worth prioritizing; management of these patients should be decided on an individualized basis. The potential benefits and toxicities of sequential or concurrent therapy with olaparib and abemaciclib are not known, and there are no data available combining a cyclin-dependent kinase 4/6 inhibitor and Poly (ADP-ribose) polymerase inhibitors. Trials are ongoing (eg, ClinicalTrials.gov identifier: [NCT03685331](https://clinicaltrials.gov/ct2/show/study?term=NCT03685331)).

Question: Should We Be Using Prophylaxis for Side Effects (eg, Diarrhea) in All Patients?

Overall, nearly 20% of participants did not complete the planned 2 years of abemaciclib in monarchE. Diarrhea was

TABLE 1. Summary of Updated Recommendations

Recommendation	Type	Evidence Quality	Strength
1.1. Based on a secondary predefined analysis conducted by the FDA, ¹¹ 2 years of abemaciclib (150 mg twice daily) plus ET may be offered to patients with HR-positive, HER2-negative, node-positive early breast cancer with a high risk of recurrence and a Ki-67 score of $\geq 20\%$ as determined by an FDA-approved test.	EB	M	S
1.2. The Panel also recommends, based on analyses reported by Harbeck et al, that abemaciclib for 2 years plus ET for ≥ 5 years may be offered to the broader ITT population of patients with resected, HR-positive, HER2-negative, node-positive, early breast cancer at high risk of recurrence, defined as having ≥ 4 positive axillary lymph nodes, or as having 1-3 positive axillary lymph nodes and one or more of the following features: histologic grade 3 disease, tumor size ≥ 5 cm, or Ki-67 index $\geq 20\%$.	EB	M	S

Qualifying statements: Although exploratory analyses suggested similar hazard ratios in favor of abemaciclib regardless of Ki-67 status, there were relatively few Ki-67 low tumors in monarchE. When discussing treatment options with patients, the potential benefits (improved IDFS) should be weighed against the potential harms (treatment toxicity and financial cost).

Abbreviations: EB, evidence-based; ET, endocrine therapy; FDA, US Food and Drug Administration; HER2, human epidermal growth factor receptor 2; HR, hormone receptor; IDFS, invasive disease-free survival; ITT, intent-to-treat; M, moderate; S, strong.

the most common toxicity observed in patients randomly assigned to abemaciclib in monarchE (all grades = 83%; grade 3 = 7.8%), and the median time to diarrhea onset is approximately 1 week.¹⁰ Antidiarrheal prophylaxis may be contraindicated because of constipation risks, and hence, early intervention in the event of symptoms is advised. As abemaciclib is initiated, patients should be educated on diarrhea management including the importance of hydration and dietary modification, with recommendations for early use of over-the-counter antidiarrheal medications (eg, loperamide) at the first sign of diarrhea. Dose modifications should be considered when diarrhea is severe, and close communication and follow-up for patients having toxicity is recommended. Given the frequency of diarrhea, in addition to nausea, neutropenia, fatigue, and other adverse events, regular monitoring with complete blood counts and electrolytes is recommended for those on abemaciclib, particularly during the early months of therapy initiation. Notably, dose adjustments (68.1%) including dose omissions (59.9%), dose reductions (41.2%), and early cessation of therapy (median duration

of abemaciclib 14 months v planned 24 months) occurred commonly in monarchE, supporting the role of dose reductions triggered by adverse events in ITT population.

Question: Is There a Preference for Which Endocrine Therapy Agent to Combine With Adjuvant Abemaciclib in Practice?

In the monarchE trial, the type of ET administered was left to the treating clinician, with 68.3% of patients receiving aromatase inhibitor (AI) therapy and 31.4% receiving tamoxifen as initial therapy, including 14.2% and 7.6%, respectively, in combination with ovarian suppression. Although the benefits of differing ET partners was not a focus of monarchE, a notably higher risk of venous thromboembolism (VTE) was observed when abemaciclib was combined with tamoxifen (any-grade VTE = 4.1%, grade ≥ 3 VTE = 2.2%) compared with AIs (any-grade VTE = 1.7%, grade ≥ 3 VTE = 0.9%). Because of these observed events, clinicians should include VTE risk assessment when deciding on a treatment regimen with patients, perhaps favoring AI use in patients receiving abemaciclib if VTE is of concern.

AFFILIATIONS

¹Dana-Farber Cancer Institute, Boston, MA

²Lifespan Cancer Institute, Providence, RI

³American Society of Clinical Oncology, Alexandria, VA

⁴Stanford University School of Medicine, Stanford, CA

⁵Sidney Kimmel Comprehensive Cancer Center at Johns Hopkins, Baltimore, MD

⁶MD Anderson Cancer Center, Houston, TX

CORRESPONDING AUTHOR

American Society of Clinical Oncology, 2318 Mill Rd, Suite 800, Alexandria, VA 22314; e-mail: guidelines@asco.org.

DISCLAIMER

No part of this document may be reproduced or transmitted in any form or by any means, electronic or mechanical, including photocopy, recording, or any information storage and retrieval system, without written permission by the American Society of Clinical Oncology.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Disclosures provided by the authors are available with this article at DOI <https://doi.org/10.1200/OP.22.00140>.

REFERENCES

1. Johnston SRD, Harbeck N, Hegg R, et al: Abemaciclib combined with endocrine therapy for the adjuvant treatment of HR+, HER2-, node-positive, high-risk, early breast cancer (monarchE). *J Clin Oncol* 38:3987-3998, 2020
2. Harbeck N, Rastogi P, Martin M, et al: Adjuvant abemaciclib combined with endocrine therapy for high-risk early breast cancer: Updated efficacy and Ki-67 analysis from the monarchE study. *Ann Oncol* 32:1571-1581, 2021
3. Harbeck N, Rastogi P, Shahir A, et al: Letter to the Editor for "Adjuvant abemaciclib combined with endocrine therapy for high-risk early breast cancer: Updated efficacy and Ki-67 analysis from the monarchE study". *Ann Oncol* 33:227-228, 2022
4. Royce M, Osgood C, Mulkey F, et al: FDA approval summary: Abemaciclib with endocrine therapy for high-risk early breast cancer. *J Clin Oncol* 40:1155-1162, 2022
5. Denduluri N, Somerfield MR, Chavez-MacGregor M, et al: Selection of optimal adjuvant chemotherapy and targeted therapy for early breast cancer: ASCO guideline update. *J Clin Oncol* 39:685-693, 2021

AUTHOR CONTRIBUTIONS

Conception and design: All authors

Administrative support: Rachel A. Freedman, Mark R. Somerfield, Sharon H. Giordano

Collection and assembly of data: All authors

Data analysis and interpretation: All authors

Manuscript writing: All authors

Final approval of manuscript: All authors

Accountable for all aspects of the work: All authors

ACKNOWLEDGMENT

Selection of Optimal Adjuvant Chemotherapy and Targeted Therapy for Early Breast Cancer: ASCO Guideline Update was developed and written by: Mariana Chavez-MacGregor, MD, MSc; Amy H. Comander, MD; Zoneddy Dayao, MD; Andrea Eisen, MD; Rachel A. Freedman, MD, MPH; Ragisha Gopalakrishnan, MD; Stephanie L. Graff, MD; Michael J. Hassett, MD, MPH; Tari A. King, MD; Gary H. Lyman, MD, MPH; Raquel Nunes, MD; Cheryl L. Perkins, MD, RPh; Melinda L. Telli, MD; Maureen E. Trudeau, MD; Antonio C. Wolff, MD; and Sharon H. Giordano, MD, MPH.

6. Giordano SH, Freedman RA, Somerfield MR, et al: Abemaciclib with endocrine therapy in the treatment of high-risk early breast cancer: ASCO optimal adjuvant chemotherapy and targeted therapy guideline rapid recommendation update. *J Clin Oncol* 40:307-309, 2022
7. Nielsen TO, Leung SCY, Rimm DL, et al: Assessment of Ki67 in breast cancer: Updated recommendations from the International Ki67 in Breast Cancer Working Group. *J Natl Cancer Inst* 113:808-819, 2021
8. Polley MY, Leung SC, Gao D, et al: An international study to increase concordance in Ki67 scoring. *Mod Pathol* 28:778-786, 2015
9. Tutt ANJ, Garber JE, Kaufman B, et al: Adjuvant olaparib for patients with BRCA1- or BRCA2-mutated breast cancer. *N Engl J Med* 384:2394-2405, 2021
10. Rugo HS, Huober J, Garcia-Saenz JA, et al: Management of abemaciclib-associated adverse events in patients with hormone receptor-positive, human epidermal growth factor receptor 2-negative advanced breast cancer: Safety analysis of MONARCH 2 and MONARCH 3. *Oncologist* 26:e53-e65, 2021
11. Highlights of Prescribing Information for Verzenio. https://www.accessdata.fda.gov/drugsatfda_docs/label/2021/208716s006s007s008lbl.pdf



AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST**Adjuvant Abemaciclib Plus Endocrine Therapy in the Treatment of High-Risk Early Breast Cancer: ASCO Guideline Rapid Recommendation Update Q and A**

The following represents disclosure information provided by authors of this manuscript. All relationships are considered compensated unless otherwise noted. Relationships are self-held unless noted. I = Immediate Family Member, Inst = My Institution. Relationships may not relate to the subject matter of this manuscript. For more information about ASCO's conflict of interest policy, please refer to www.asco.org/rwc or ascopubs.org/op/authors/author-center.

Open Payments is a public database containing information reported by companies about payments made to US-licensed physicians ([Open Payments](#)).

Rachel A. Freedman

Employment: Firefly Health (I)

Stock and Other Ownership Interests: Firefly Health (I)

Research Funding: Puma Biotechnology (Inst), Eisai (Inst), Genentech/Roche (Inst)

Stephanie L. Graff

Stock and Other Ownership Interests: HCA Healthcare

Honoraria: Publicis Health

Consulting or Advisory Role: Seattle Genetics (Inst), Novartis (Inst), Pfizer, Daiichi Sankyo/Lilly, AstraZeneca, Genentech

Speakers' Bureau: OncLive

Research Funding: Boehringer Ingelheim (Inst), Lilly (Inst), Genentech (Inst), Immunomedics (Inst), Novartis (Inst), Celldex (Inst), Dana Farber Cancer Hospital (Inst), TapImmune Inc (Inst), Merus NV (Inst), Odonate Therapeutics (Inst), Innocrin Pharma (Inst), GRAIL (Inst), AstraZeneca (Inst), Bristol Myers Squibb (Inst), Daiichi Sankyo (Inst), Eisai (Inst), Roche (Inst), H3 Biomedicine (Inst), Merck (Inst), Foundation Medicine (Inst), Seattle Genetics (Inst), Taiho Pharmaceutical (Inst), Sermonix Pharmaceuticals (Inst), Polyphor (Inst)

Melinda L. Telli

Consulting or Advisory Role: Merck, Immunomedics, Genentech/Roche, G1 Therapeutics, Natera, Pfizer, OncoSec, Blueprint Medicines, Guardant Health, Novartis, AstraZeneca, Sanofi, Reflexion Medical, Gilead Sciences

Research Funding: Novartis (Inst), PharmaMar (Inst), AbbVie (Inst), Calithera Biosciences (Inst), Genentech (Inst), Medivation (Inst), OncoSec (Inst), Vertex (Inst), Biothera (Inst), Tesaro (Inst), Pfizer (Inst), EMD Serono (Inst), Bayer (Inst)

Other Relationship: G1 Therapeutics, Immunomedics

Antonio C. Wolff

Consulting or Advisory Role: Ionis Pharmaceuticals

Patents, Royalties, Other Intellectual Property: Antonio Wolff has been named as inventor on one or more issued patents or pending patent applications relating to methylation in breast cancer, and has assigned his rights to JHU, and participates in a royalty sharing agreement with JHU

Open Payments Link: <https://openpaymentsdata.cms.gov/physician/357301/summary>

No other potential conflicts of interest were reported.